The landscape of neoantigen in Korean NSCLC patients

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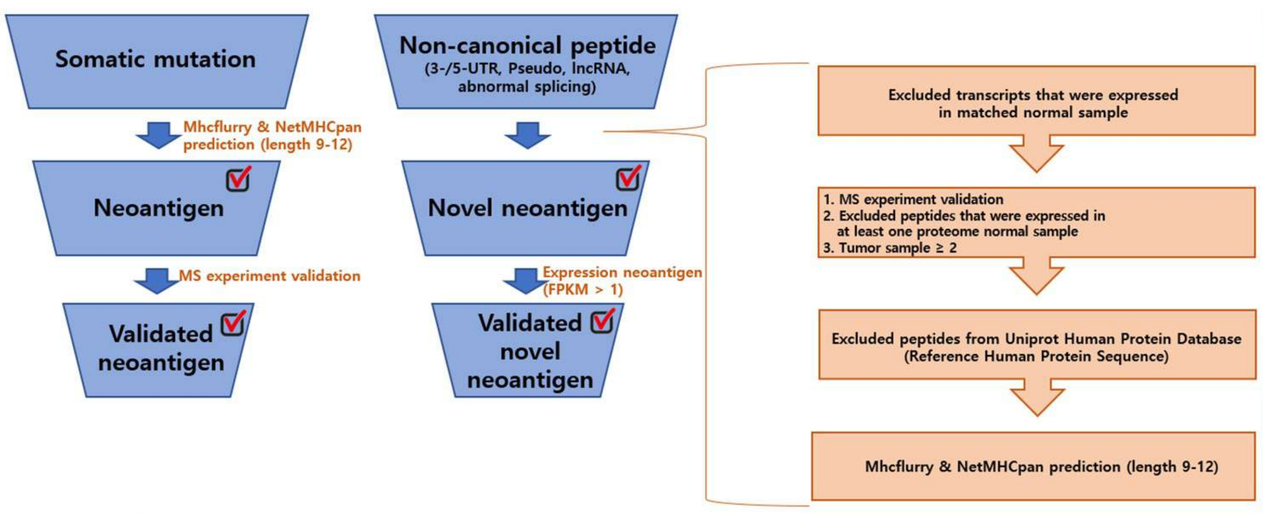
Introduction

Innate immunity and adaptive immune responses are invoked when an unrecognized virus enters the body or neoantigens from a cancer cell. The ability to induce adaptive immune responses largely depends on antigen presentation through human leukocyte antigen (HLA) class I and II molecules[1]. HLA molecules are known to be involved in the antigen presentation process of the acquired immune system and is a representative immune-related factor. HLA genes encode a cell membrane glycoprotein molecule expressed on the surface of a human nucleated cell that binds to an antigen inside the cell and then presents it to the T lymphocytes, causing an adaptive immune response to the antigen[3]. Peptides that are unusual in cancer cells are presented to T cells through MHC molecules are an important mechanism for activating adaptive immunity. Cancer patients who have high levels of neoantigens that can induce immunity may have a better survival rate than those who do not[9]. Most of the peptides that act as neoantigens in our body are from cancer mutations, and these are called canonical peptides. However, 3-/5-UTR, pseudo, lncRNA, and non-canonical peptide derived from abnormal splicing, which is not transcribed in normal cells, can also act as neoantigens[4]. I hypothesized that neoantigens were favorably associated with survival in NSLCL patients. In addition, We analyze neoantigen counts and the expression level of the antigen-presenting pathway together, and this result was assumed to be different for each multi-omics subtype.0

Methods

Identification of novel neoantigens

We classified neoantigens into 4 types.



1. Neoantigen
We found the somatic mutation region based on genomic data. We made peptide fragments containing a mutation region to predict the HLA allele and binding for each patient, and each peptide fragment was made in 9-12 lengths. Also, we typed HLA for each patient. HLA typing was based on WGS data using OptiType[5]. Binding affinity to HLA class I molecules was calculated for all lengths 9-12 peptides from detected proteins using the MHCflurry v2.0[6] and NetMHCpan v4.1[7] software. In MHCflurry, the only peptide with a binding affinity is less than 500nM was decided to bind with HLA. In NetMHCpan, strong and weak binder peptides (%Rank <0.5 and %Rank < 2%) were kept for further analysis, and we named them 'neoantigen'.

2. Validated neoantigen
We validated neoantigen by MS experiment proteomic data, and we named them 'validated neoantigen'.

3. Novel neoantigen
We identified novel peptides as 3-/5-UTR, Pseudo, lncRNA peptides by abnormal transcription and novel transcript by abnormal splicing (discovered by StringTie[8] using transcriptomic data). And we only used transcripts that were not expressed in the matched normal sample out of these non-canonical transcripts. After this, we validated transcript by MS experiment proteomic data, and we excluded peptides that were expressed in at least one proteome normal sample. For excluding experiment noise, we only used transcripts that were found in more than two tumor samples. Similarly, we made peptide fragments with transcripts in 8-12 lengths. To exclude peptides that may be generated from the reference human protein sequence, we used Uniprot, RefSeq and/or Gencode human protein sequences. Finally, we predicted binding affinity and decided on neoantigens, and we named them 'novel neoantigen'.

4. Validated novel neoantigen
In order to see only the neoantigen sufficiently expressed, we validated 'novel neoantigen' by FPKM more than 1, and we named them 'validated novel neoantigen'.

Reference

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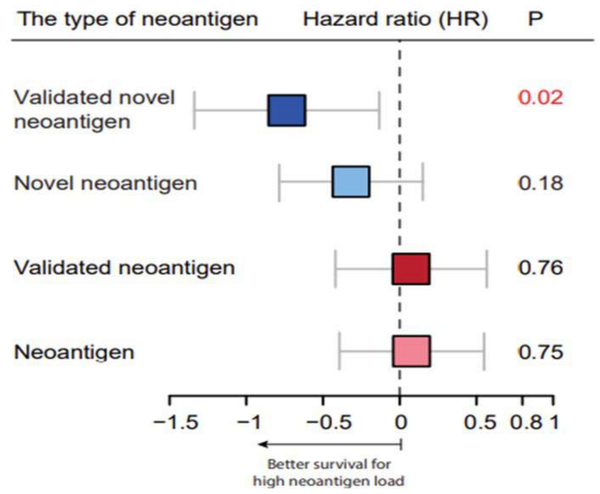
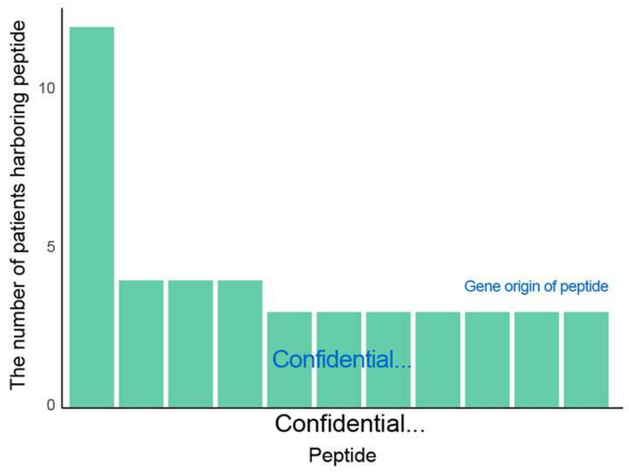
Results

Association between novel neoantigen and survival prognosis

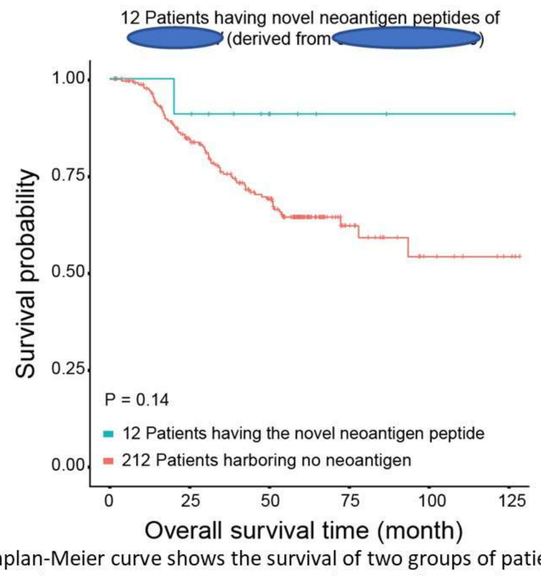
Neoantigen is another key factor affecting immune landscape in cancer patients[9]. Therefore, I identified neoantigens in our Korean NSCLC cohort using multi-omics dataset. We divides neoantigens in four subtypes, and all the details of discovery procedures for each neoantigen were described in Methods.
Including novel neoantigens, We tested how the four types of neoantigens were associated with patient survival. Interestingly, validated novel neoantigens had a positive correlation with favorable survival most significantly although the number of neoantigens was considerably low across patients (Figure 6A, B). We also found that 11 peptides of validated novel neoantigens were observed in more than three patients and some of them were derived from an identical origin of a gene (Figure 6C). Furthermore, the load of top recurrent neoantigen was correlated with a longer survival although it was nonsignificant (Figure 6D). These results suggest a neoantigen that occurs commonly in more than 5% of NSCLC patients, and this neoantigen can be considered as a cancer vaccine candidate.



> The distribution of the neoantigen load for each patient was described accordingly by the type of neoantigen



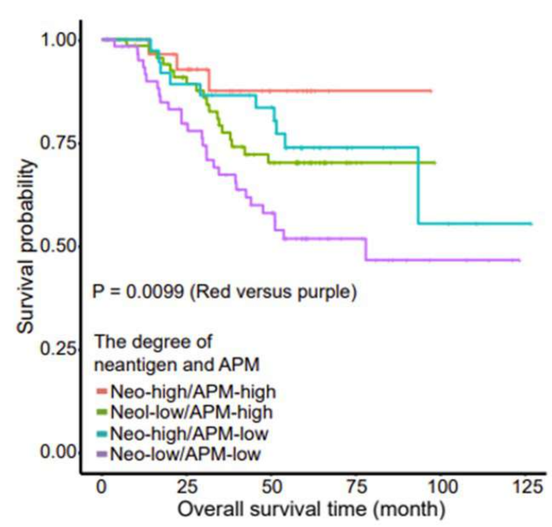
> Bar plot shows the number of neoantigen peptides observed in more than 3 patients in descending order.



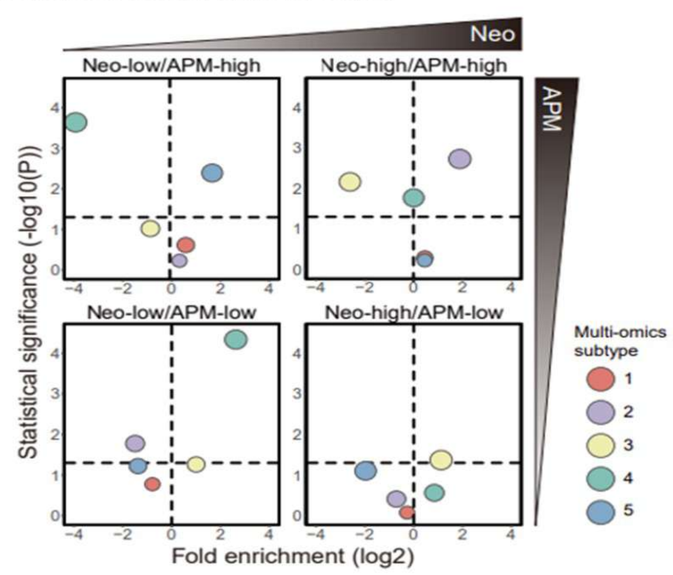
> Kaplan-Meier curve shows the survival of two groups of patients (n = 224) divided by whether they have (blue line) or not (red line) the neoantigen peptide that was **confidence**. This peptide was derived from **confidence**.

Multi-omics profiling of neoantigen along with APM

We next investigated how the novel neoantigens were associated with antigen processing and presentation machinery (APM) and multi-omics subtype. APM has been proposed as a critical factor to evaluate immune activating potential along with neoantigens in the previous study[10]. We hypothesized that neoantigen was associated with multi-omics subtypes when it was considered together in the context of immune landscape. In our cohort, the overall patient prognosis was stratified by adding APM to neoantigen load. Patients having high neoantigen load and activated APM showed the longest survival, in contrast to the worst survival of the patients with low neoantigen load and deactivated APM. We then calculated the distribution of multi-omics subtypes along with the degree of neoantigen load and APM together. Multi-omics subtype 2 was most associated with high neoantigen load and deactivated APM, whereas subtype 5 was most enriched in low neoantigen load and activated APM. Specifically, multi-omics subtype 4 showed specific enrichment in both low neoantigen load and APM. It could be interpreted that subtype 2 may evoke a strong immune activation with more neoantigens that are presented, and therefore showed the best prognosis than other subtypes. However, patients of subtype 4 had less neoantigens with repressed presenting machinery showed the worst prognosis. Patients of subtype 5 that showed moderate survival had low neoantigen load with activated APM. In this situation, the moderate survival of this subtype seems to be associated with low neoantigen load despite of an active immune landscape. These results imply that understand of patient prognosis and multi-omics subtype needs multifactorial consideration based on TIME.



> Kaplan-Meier curve shows the survival pattern of the four groups of patients (n = 174) divided by neoantigen load and APM.



> X, Y-axis indicates an enrichment and a statistical significance calculated using two-sided Fisher's exact test with Benjamini-Hochberg adjustment. The size of the dot represents the statistical significance.